

# B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program  
Subject – Data Science with AIML

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## Topic – Basic Python | Assignment: 2

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### Question 1 – Tuple: my\_tuple

Create a tuple and print the first element, last element, length, and elements from index 1 to 3.

```
my_tuple = (10, 20, 30, 40, 50)

print("First element :", my_tuple[0])      # 10
print("Last element  :", my_tuple[-1])     # 50
print("Length        :", len(my_tuple))    # 5
print("Index 1 to 3   :", my_tuple[1:4])   # (20, 30, 40)
```

**Output:**

```
First element : 10
Last element  : 50
Length       : 5
Index 1 to 3  : (20, 30, 40)
```

### Question 2 – Tuple: fruits

Create a tuple and print the second fruit, last two fruits, and total number of fruits.

```
fruits = ("apple", "banana", "mango", "orange")

print("Second fruit      :", fruits[1])      # banana
print("Last two fruits   :", fruits[-2:])    # ('mango', 'orange')
print("Total number      :", len(fruits))    # 4
```

**Output:**

```
Second fruit      : banana
Last two fruits   : ('mango', 'orange')
Total number      : 4
```

### Question 3 – Set: numbers

Create a set and print the complete set, its length, and check if 30 is present.

```
numbers = {10, 20, 30, 40, 50}

print("Complete set :", numbers)
print("Length       :", len(numbers))      # 5
print("30 present?  :", 30 in numbers)     # True
```

**Output:**

```
Complete set : {10, 20, 30, 40, 50}
Length      : 5
30 present? : True
```

### Question 4 – Set Operations: Union & Intersection

Create two sets and find their union and intersection.

```
set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}
```

```
print("Union          :", set1 | set2)    # {1, 2, 3, 4, 5, 6}
print("Intersection :", set1 & set2)    # {3, 4}
```

#### Output:

```
Union          : {1, 2, 3, 4, 5, 6}
Intersection : {3, 4}
```

### Question 5 – Dictionary & List: student + largest/smallest

Create a student dictionary and print its fields. Then find the largest, second largest, and smallest element from a list.

```
student = {"name": "Kriti", "age": 20, "course": "Python"}

print("Complete dictionary :", student)
print("Student name       :", student["name"])
print("Student age        :", student["age"])
print("Student course     :", student["course"])

# List operations
numbers = [12, 45, 7, 23, 56, 89, 34]
sorted_numbers = sorted(numbers, reverse=True)

print("\nLargest element       :", sorted_numbers[0])    # 89
print("Second largest element :", sorted_numbers[1])    # 56
print("Smallest element      :", sorted_numbers[-1])    # 7
```

#### Output:

```
Complete dictionary : {'name': 'Kriti', 'age': 20, 'course': 'Python'}
Student name       : Kriti
Student age        : 20
Student course     : Python

Largest element       : 89
Second largest element : 56
Smallest element      : 7
```

### Question 6 – Reverse a List Without reverse()

Write a function that returns a list in reverse order without using the built-in reverse() method.

```
def reverse_list(lst):
    reversed_lst = []
    for i in range(len(lst) - 1, -1, -1):
        reversed_lst.append(lst[i])
    return reversed_lst

arr = [10, 20, 30, 40, 50, 60]
print("Original list :", arr)
print("Reversed list :", reverse_list(arr))
```

#### Output:

```
Original list : [10, 20, 30, 40, 50, 60]
Reversed list : [60, 50, 40, 30, 20, 10]
```

### Question 7 – Tuple: data (divisible by 5, sum, average)

Count elements divisible by 5, find the sum, and calculate the average.

```
data = (5, 10, 15, 20, 25, 30, 35)
```

```

count_div5 = sum(1 for x in data if x % 5 == 0)
total      = sum(data)
average    = total / len(data)

print("Divisible by 5 :", count_div5)    # 7
print("Sum           :", total)          # 140
print("Average       :", average)        # 20.0

```

#### Output:

```

Divisible by 5 : 7
Sum           : 140
Average       : 20.0

```

### Question 8 – Dictionary: Highest/Lowest Marks & Filter

Find the student with highest marks, lowest marks, and list students who scored above 85.

```

students = {
    "Aman" : 78,
    "Riya" : 92,
    "Kriti": 88,
    "Rahul": 95
}

highest = max(students, key=students.get)
lowest  = min(students, key=students.get)

print("Highest marks :", highest, "->", students[highest])
print("Lowest marks  :", lowest, "->", students[lowest])

print("Students with more than 85 marks:")
for name, marks in students.items():
    if marks > 85:
        print(" ", name, "->", marks)

```

#### Output:

```

Highest marks : Rahul -> 95
Lowest marks  : Aman  -> 78
Students with more than 85 marks:
Riya  -> 92
Kriti -> 88
Rahul -> 95

```

### Question 9 – count\_frequency(arr)

Write a function that prints the frequency of each element in a list.

```

def count_frequency(arr):
    frequency = {}
    for item in arr:
        frequency[item] = frequency.get(item, 0) + 1
    for key, count in sorted(frequency.items()):
        times = "time" if count == 1 else "times"
        print(f"{key} -> {count} {times}")

arr = [1, 2, 2, 3, 1, 4, 2]
count_frequency(arr)

```

#### Output:

```
1 -> 2 times
2 -> 3 times
3 -> 1 time
4 -> 1 time
```

### Question 10 – find\_duplicates(arr)

Write a function that prints all duplicate elements present in a list.

```
def find_duplicates(arr):
    seen = set()
    duplicates = set()
    for item in arr:
        if item in seen:
            duplicates.add(item)
        else:
            seen.add(item)
    print("Duplicate elements are:")
    for item in sorted(duplicates):
        print(item)

arr = [10, 20, 30, 20, 40, 10, 50, 30]
find_duplicates(arr)
```

#### Output:

```
Duplicate elements are:
10
20
30
```